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Development and Application of an AI Visual Defect Detection System for Warp-Knitted Lace Based on 5G+ Technology

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Abstract

Conventional defect inspection for warp-knitted lace relies on manual work and negative-sample-based training, resulting in low efficiency, frequent false detections and poor adaptability. This study presents a novel AI visual inspection system centered on positive-sample learning, which is built upon a five-layer 5G + Industrial Internet distributed architecture. Supported by modified looms, high-precision imaging devices and an optimized YOLOv5s model, the system accomplishes intelligent defect detection. A positive-sample self-learning paradigm and dual-model collaboration mechanism are proposed to reduce the demand for negative samples and cut labeling expenses. The integration of CBAM, FPN + PAN structure, self-supervised learning and hybrid loss further strengthens the recognition performance for subtle defects under complex patterns. Industrial tests show that the system reaches a grid-level classification accuracy of 95% and a frame-level detection rate over 98%, with a detection speed of 30 m/min. It reduces labor costs and product reject rates by 40% and 30% correspondingly while running stably in real production. This method breaks the constraints of traditional training modes, provides a scalable intelligent solution for the digital upgrading of the warp-knitted lace industry, and promotes the high-quality development of textile manufacturing.

Keywords: 5G+ technology; warp knitting lace; AI visual inspection; positive sample self-learning; machine vision

1. Introduction

Textile manufacturing is a pillar traditional industry in Fujian Province, China, accounting for nearly one-fifth of the region's total manufacturing enterprises. Centered in Changle District, the local textile sector has developed into a hundred-billion-yuan industrial cluster with a complete industrial chain covering petrochemical raw materials, chemical fiber production, cotton spinning, knitting, dyeing and finishing, garment making and textile machinery manufacturing. Over time, Changle has built three world-leading industrial bases. Firstly, it hosts the world's largest nylon filament production base with an annual output of 1.5 million tons, occupying over one-third of China's total production. Yongrong Group and Hengshen Group rank first and second nationwide in nylon filament output, and the local caprolactam production base also takes the leading position globally. Secondly, Changle is China's largest production hub for warp-knitted lace, with an annual capacity of nearly 1 million tons, accounting for 60% of the national total. The district is

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equipped with 15,000 warp knitting machines and develops more than one million lace patterns, leading the country in both equipment scale and pattern diversity. Thirdly, it possesses China's largest chemical fiber blended yarn base with an annual output exceeding 3 million tons, making up 50% of the national share. The local cotton spinning spindles exceed 7 million units, ranking first among all county-level regions across China.

Driven by the ongoing technological revolution and industrial transformation, digital upgrading has become an irresistible trend for traditional industries [1]. To achieve high-quality development, textile clusters need to embrace digital and intelligent transformation via next-generation information technologies. As a high-precision and high-value-added segment of textile production, warp-knitted lace manufacturing sets extremely strict requirements for quality inspection. Traditional fabric defect detection mainly adopts manually designed rules and conventional image processing techniques, which generally consist of two core procedures: feature extraction and defect identification [2–5]. Feature extraction is a fundamental step that extracts representative visual characteristics from raw images to describe defective regions and generate quantifiable inputs for subsequent classifiers and decision rules. Based on the regularity of pixel distribution, this process converts heterogeneous visual information into structured data. The performance of defect detection is largely determined by the representation ability of extracted features, including geometric, textural and frequency-domain features.

Conventional detection methods can be categorized into three types: statistical analysis, frequency-domain analysis and model-based approaches [3,6]. Statistical methods distinguish defects by comparing grayscale and texture differences between normal and defective areas, with typical implementations including grayscale histograms, gray-level co-occurrence matrices (GLCM) and mathematical morphology. Frequency-domain methods identify anomalies by analyzing the spectral periodicity and directionality of fabric textures, represented by Fourier transform, wavelet transform and Gabor filters. Model-based methods establish standard defect-free templates and locate defects by calculating the deviation between test images and templates, such as Markov random fields and autoregressive models. Traditional methods have clear implementation paths in image preprocessing, feature extraction and classification. With manually crafted features and rules, they can work well for specific fabric types and common defects under limited data and computing resources. Nevertheless, the continuous diversification of fabric varieties and defect types leads to increasingly complex texture features, which greatly limits the representation capability, adaptability and robustness of conventional algorithms [4,6,7].

This problem is particularly prominent for warp-knitted lace, which features intricate structures, diverse mesh styles and three-dimensional surfaces, making traditional detection methods ineffective [6,8]. Common defects such as broken yarns, dropped stitches, fly yarns, misaligned yarns and pattern errors vary greatly in appearance. Some defects share highly similar visual features with normal textures, further increasing detection difficulty. In addition, conventional learning methods rely heavily on large-scale defective samples for model training, which restricts the detection accuracy and generalization performance [9]. Current machine vision-based online inspection solutions mostly depend on two-dimensional image analysis. They fail to fully characterize the multi-layer and multi-dimensional structures of lace, resulting in a high false detection rate [2,4]. Differentiating inherent texture patterns from real defects has become a core technical bottleneck for lace inspection, which puts forward higher requirements for visual algorithms. Meanwhile, the huge variety and structural differences of lace products also hinder the cross-factory application of existing detection systems.

For decades, warp-knitted lace manufacturers have relied on manual inspection due to its complex textures and three-dimensional forms. Manual detection brings frequent prod-

uct returns and rework, restricting the optimization of production design, manufacturing processes, quality control and cost reduction. To solve this widespread industrial problem, developing advanced AI vision inspection technologies has become an inevitable direction for industrial innovation [8,10,11]. Against this background, this study proposes an intelligent inspection scheme for warp-knitted lace. Different from conventional solutions that depend on massive defective samples, the proposed method adopts a positive-sample self-learning training strategy to improve detection accuracy and adaptability. Targeting the difficulties in automatic identification and classification of lace surface defects, this paper optimizes machine vision algorithms and edge-cloud collaborative deployment architectures, and finally presents a high-efficiency intelligent detection solution for warp-knitted lace production [12–14].

2. Overall System Architecture

2.1. Overall System Design Concept

This system adopts 5G + Industrial Internet as its core technical framework and establishes an intelligent quality inspection and data management system covering the whole process of warp-knitted lace manufacturing [12,13,15]. The overall design follows a five-layer distributed architecture, device layer—network layer—AI layer—application layer—management layer, forming an intelligent closed loop of perception–computing–decision–execution–feedback [14,16]. In this architecture, the device layer is responsible for image acquisition and state signal sensing; the network layer realizes high-speed communication and edge computing; the AI layer completes algorithm inference and model training; the application layer undertakes human–computer interaction and visualization tasks; and the management layer implements policy scheduling, model updating and authority control.

2.2. System Structure and Hierarchical Functions

The device layer consists of knitting machines, industrial cameras, light sources, sensors, motor controllers and other equipment. It collects multi-dimensional data in real time, including fabric surface images, machine start–stop signals, speed, density, tension and so on, forming a digital mapping of fabric status. The network layer constructs a low-latency communication channel based on a 5G private network and deploys Multi-Access Edge Computing (MEC) nodes to realize edge-side inference and rapid response. The communication latency of the system is controlled below 30 ms, meeting the requirements of real-time quality inspection. The AI layer contains four sub-modules: data sampling, algorithm training, model inference and self-learning [9,17,18]. The model adopts a multi-scale convolutional neural network structure integrated with an attention mechanism to improve the accuracy of feature recognition. The application layer includes client and server sides. The client is used for operation control and defect review; the server is responsible for data statistics, report analysis and process optimization decision-making. The management layer provides functions such as algorithm version control, equipment monitoring, user authority management and quality traceability, ensuring the security and maintainability of long-term system operation.

To satisfy the real-time demands of high-speed industrial production, we adopted a collaborative architecture of 5G private network and Multi-Access Edge Computing (MEC). This solution enables low-latency transmission of fabric images, machine status data and control signals, and supports edge-side model inference as well as remote device control, effectively guaranteeing the real-time performance of online defect detection. We conducted a latency performance comparison between the 5G-based scheme and the traditional wired network. Experimental results indicate that the 5G private network reduces the network

transmission latency to no more than 30 ms, which is a significant improvement over the 80–120 ms of the traditional wired solution. For the end-to-end inference process, the overall latency from image input to detection result output is controlled within 50 ms. In terms of defect response speed, the latency from defect alarm triggering to loom stopping is compressed to $\leq$45 ms, far outperforming the traditional scheme which requires 100–180 ms for the same process. These results fully verify the efficiency of the 5G-MEC architecture in supporting real-time industrial inspection.

3. Technical Scheme

3.1. Knitting Machine Retrofit

Warp knitting machines are characterized by high-speed operation, strong vibration, and compact structure. To achieve stable imaging and precise control, this project carries out modular retrofitting of the machine structure, as detailed in Figure 1 (Design Scheme of Warp Knitting Machine Retrofit) and Figure 2 (On-site Effect). Shock-proof brackets and polymer damping pads are installed in the imaging area to reduce image blur caused by camera vibration [19]. Aiming at the differences between RSJ and RSE series knitting machines (Karl Mayer Textilmaschinen GmbH, Obertshausen, Germany), an adjustable frame is designed to support the adjustment of the shooting angle within a range of 10° to 20°. The distance between the camera and the knitting line is maintained at 300–350 mm, with a shooting area of 0.1 m^2, ensuring uniform image resolution. The light source bracket is made of aluminum alloy to prevent electromagnetic interference and improve heat dissipation performance.

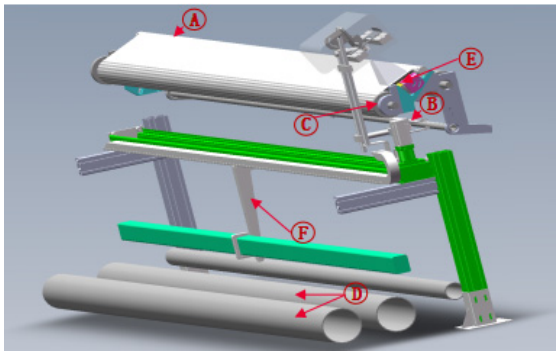


Figure 1. Design Scheme of Warp Knitting Machine Retrofit.



Figure 2. On-site Effect.

Intelligent capabilities are constructed around five operating scenarios, including AI inspection, shutdown processing, defect archiving, doffing, and shift output, through the integrated retrofitting of machinery, optics, electricity, computing, and software. The main components are as follows:

A. Industrial camera: a 5-megapixel area scan camera with hardware trigger function. Industrial lens: C-mount fixed-focus lens (Fujinon, Saitama, Japan), focal length 6 mm, aperture F1.4–22, sensor format 1/2.5 inch. Considering fabric vibration during weaving, the lens depth of field is required to reach 10 mm.
B. All-in-one computer: configured with Y8161A5 mainboard (Advantech Co., Ltd., Taipei, Taiwan, China), 8 GB RAM, 256 GB SSD, Windows 10 64-bit (Version 22H2, Microsoft Corporation, Redmond, WA, USA), built-in Wi-Fi, and capacitive touch screen. Functions: communicate with PLC to implement equipment control; send captured images to the algorithm processing system; and receive and display algorithm processing results.
C. Servo motor (Mitsubishi Electric, Tokyo, Japan): power 400 W, torque 1.25 N·m. Reducer (Shimpo Drives, Kyoto, Japan): when assembled with the servo motor, the maximum speed reaches 1 m/s. PLC (Siemens AG, Munich, Germany): connected with the knitting machine running signal, stop signal, photoelectric switch signal, and meter counter signal, and communicates with the computer via RS485 to realize software control of the LED light source.
D. Mounting mechanism: connects the camera and light source to the moving platform, ensuring adjustable angles relative to the fabric surface. Flexible cables: 1 flexible Ethernet cable and 2 flexible power cables.
E. Light source: 2 LED strip light sources (Keyence Corporation, Osaka, Japan), length 250 mm, width 50 mm, with adjustable brightness.
F. Moving guide rail (Dongguan Ruida Automation Equipment Co., Ltd., Dongguan, Guangdong, China): Maximum stroke 3500 mm (customized according to on-site machines), load 3 kg; maximum running speed 1 m/s, adjustable in real time according to fabric movement speed to ensure 5–10 mm vertical overlap between consecutive images. A trigger signal is sent to capture one image every 100 mm (adjustable) based on the camera field of view, so as to guarantee 5–10 mm overlap between adjacent frames.

3.2. Photoelectric System Design

The photoelectric system constitutes the critical input module for AI-based defect detection [20]. The configuration of the knitting machine side is presented in Table 1. A Basler acA2440-35gm industrial camera (Basler AG, Ahrensburg, Germany) (resolution: 2448 × 2048, frame rate: 35 fps) is adopted, which cooperates with a linear sliding guide to achieve a 20 s reciprocating scan, delivering an imaging precision of 0.1 mm. The illumination system adopts a combination of bilateral LED strip lights and backlight transmission lighting, with a side light incident angle of $40^\circ \pm 5^\circ$ to enhance the contrast of fiber undulation, and the schematic diagram of the photoelectric system design is shown in Figure 3. Light intensity is regulated by a PWM automatic adjustment module, enabling the system to dynamically adjust exposure according to fabric surface reflectivity [21,22].

The system features high precision and high real-time performance in fabric image acquisition and machine data synchronization. To achieve full-coverage, blind-zone-free imaging, the knitting machine side uses a single area-scan camera coupled with mechanical reciprocating motion to scan the entire fabric surface periodically. The camera completes full-surface imaging within a 20 s reciprocating cycle, ensuring the integrity and continuity of fabric image data during weaving. In the fabric inspection stage, the system is equipped with five fixed area-scan cameras that capture images synchronously at 4 frames per second, corresponding to an inspection speed of approximately 24 m/min, thus enabling efficient and full-coverage imaging of moving fabrics. In terms of imaging accuracy, the system achieves a resolution of 0.1 mm, allowing precise detection and identification of anomalies

in approximately 90 d single yarns, meeting high-precision requirements such as weaving defect detection, surface texture analysis, and intelligent recognition.

Table 1. Key Parameters of the Photoelectric System.

Item	Configuration Parameters	Function Description
Camera Model	Basler acA2440-35gm/IDS UI-3280CP(Imaging Development Systems GmbH, Obersulm, Germany)	Area-array imaging
Light Source Type	LED strip light + backlight transmission light	Enhance fabric surface detail contrast
Shooting Angle	35°~45°	Reduce specular reflection
Imaging Precision	0.1 mm	Capture anomalies of 90 d single yarn
Fabric Coverage	100%	Full-width scanning without blind zones

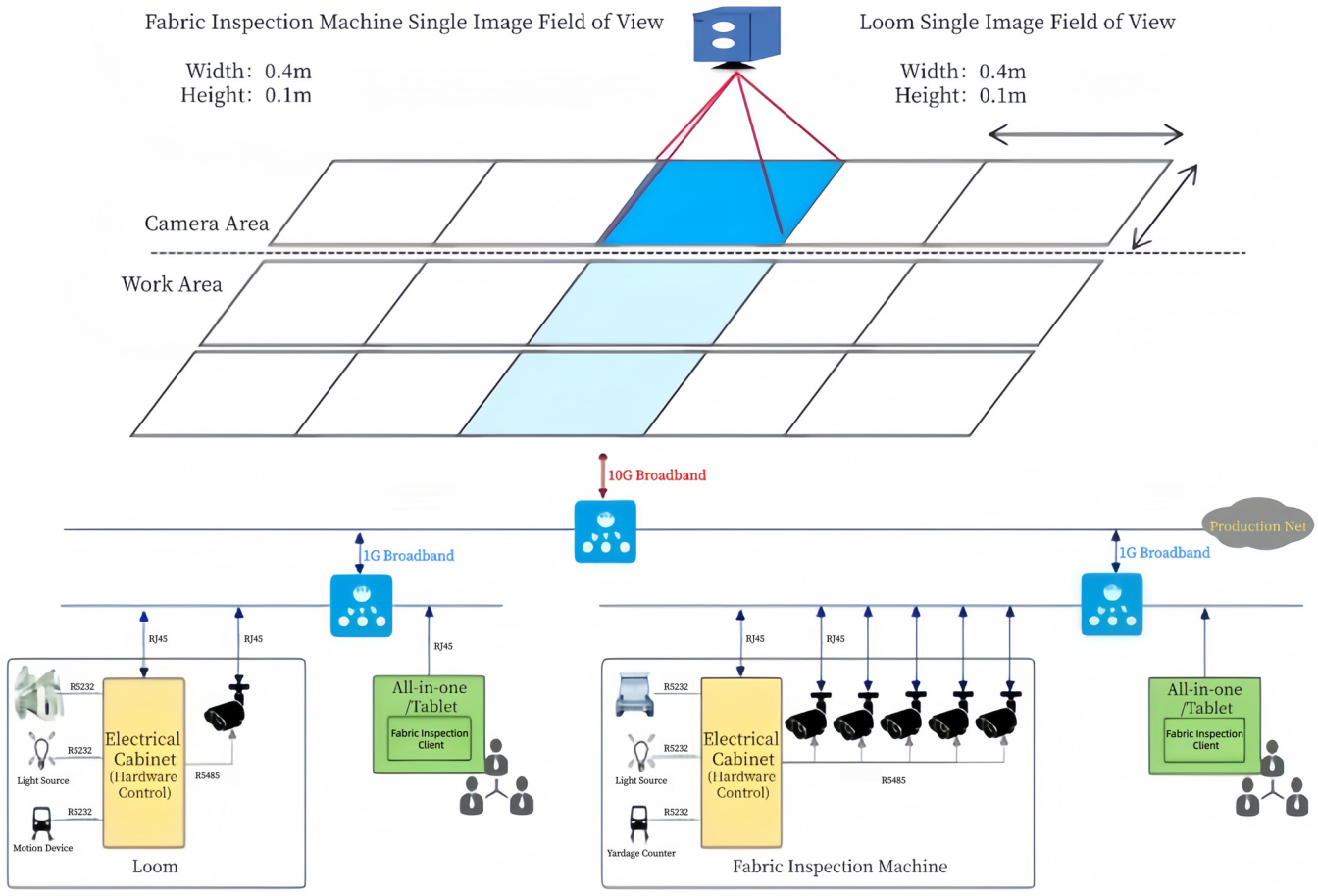


Figure 3. Schematic Diagram of Photoelectric System Design.

In addition, the system supports flexible control of five types of devices, including knitting machines, signal lights, light sources, motors, and cameras, enabling remote start–stop and dynamic adjustment based on process parameters and operating conditions. To ensure data timeliness, the system reports machine operating status (including run–stop state, rotating speed, meter counter value, etc.) at the second level with a sampling interval of 1 s, realizing real-time synchronization and precise correlation between machine data

and image data. This photoelectric acquisition framework provides a high-precision, high-reliability data foundation for fabric quality monitoring and data fusion in intelligent weaving.

3.3. AI Algorithm System

3.3.1. Data Sampling and Preprocessing

The system acquires fabric surface images through continuous shooting by a slide-mounted camera, collecting approximately 300 frames every 20 s. The images are processed via automatic blocking (128 × 128 pixel grids), gamma correction, and brightness equalization before entering the data augmentation stage. This system adopts a semi-automatic label generation scheme to reduce manual annotation workload, which eliminates the need for full manual bounding box labeling across all samples. For normal lace samples, no manual annotation is required, and these images can be directly applied to self-supervised learning tasks. As for defective samples, only a small number of typical defective images need simple manual verification, while most labels are produced automatically. To generate standard YOLO-format labels, technicians first manually annotate 5 to 10 representative samples for each defect category with YOLO bounding boxes containing category information and normalized coordinate parameters, which serve as seed labels. The pre-trained positive model is then utilized to extract defect feature templates. By conducting similarity matching on unlabeled defective images, the system automatically generates pseudo bounding boxes and category labels in accordance with YOLO specifications. An error tolerance rate of no more than 5% is configured for defective samples to filter out low-confidence pseudo labels and preserve high-quality data for model training. Furthermore, manually revised results are fed back to update the pseudo-label templates during the self-learning process, establishing a closed-loop mechanism for continuous label optimization. All labels used for model training comply with the standard YOLOv5 format, which consists of normalized coordinates in the form of [class_id, x_center, y_center, width, height]. Augmentation operations include random rotation within $\pm 10^{\circ}$, Gaussian noise (σ = 0.01), brightness adjustment within $\pm$20%, and contrast modification within $\pm$15%, which effectively improve the robustness of the model [17,18,23].

Positive-Sample Self-Learning Method

This study develops a Positive-Sample Self-Learning (PSSL) framework dedicated to defect detection for warp-knitted lace. Traditional defect detection techniques rely heavily on a large volume of defective samples to complete model training, whereas the proposed framework takes defect-free lace images as the primary training dataset. This method adopts self-supervised learning to characterize the inherent distribution of texture, pattern and structural features of standard warp-knitted lace, and realizes anomaly discrimination by quantifying the feature deviation between newly acquired images and the pre-established normal feature space. Since only a limited number of defective samples are introduced for auxiliary calibration, the proposed methodology effectively alleviates the dependence on labor-intensive manual data annotation.

Let $\boldsymbol{X}_p = x_1, x_2, \ldots, x_N$ denote the set of normal lace samples, and $F(\cdot;\theta)$ represent the feature extractor with trainable parameters θ. The self-reconstruction loss function is formulated as $\mathrm{L}_{self} = \frac{1}{N}\sum_{i=1}^{N}\left|F(x_i;\theta) - \hat{F}(x_i;\theta)\right|_2^2$, where $\hat{F}(\cdot)$ refers to the feature reconstruction branch embedded in the network. For an arbitrary input image x, the L2 distance between its extracted features and the centroid of normal features $\boldsymbol{\mu}_p = \frac{1}{N}\sum_{i=1}^{N} F(x_i;\theta)$ is calculated as $D(x) = \left|F(x;\theta) - \boldsymbol{\mu}_p\right|_2$. A predefined threshold τ is adopted for classification: an input sample is identified as normal if $D(x) \leq \tau$, and is judged to be defective if $D(x) > \tau$. Combined with a small-scale defective sample set $\boldsymbol{X}_n = y_1, y_2, \ldots, y_M$ satisfying $M \ll N$,

the overall loss function is defined as $L_{total} = L_{self} + \lambda \cdot L_{cls}(\boldsymbol{X}_p, \boldsymbol{X}_n)$. In this formulation, the weighting coefficient λ is set to 0.2 throughout the experiments, and L_{cls} acts as the classification loss for few-shot defect calibration.

The implementation procedure of the PSSL framework is elaborated as follows. To begin with, all model parameters are initialized, and the feature extractor is trained using pure normal samples by minimizing the self-reconstruction loss. The centroid of normal feature distribution is then computed after the initial training phase. Subsequently, the model is fine-tuned on the mixed dataset containing both normal and defective samples via the joint loss function. In the inference stage, the final detection results are determined by comparing the calculated feature distance with the preset threshold.

The overall operating workflow follows a sequential execution process. Firstly, unlabeled normal lace images are collected and preprocessed to train the feature extractor based on self-supervised reconstruction. After the centroid of the normal feature space is obtained, few-shot fine-tuning is implemented with sparsely annotated defective samples. In practical industrial deployment, the system accomplishes anomaly detection by measuring the feature deviation of real-time captured images.

Dual-Model Collaboration Mechanism

This system implements a dual collaborative framework consisting of a positive model and a negative model, which is well adapted to the proposed positive-sample self-learning paradigm. Both models adopt the improved YOLOv5s network embedded with the CBAM attention module and FPN + PAN multi-scale feature fusion structure as the shared backbone, while they are assigned distinct tasks, training targets and feature learning priorities. The detection results from the two models are fused during the inference phase to reduce misjudgments induced by intricate lace patterns and inconspicuous subtle defects.

The positive model shares the same backbone architecture as the overall detection network and occupies a model size of 85 MB. It is trained primarily on normal lace samples mixed with 5% defective samples, and optimized with self-reconstruction loss and BCE loss. This model aims to learn the standard feature distribution of diverse normal lace patterns, remove recognition bias among different normal textures, and distinguish normal texture regions from potential defective areas. Its output corresponds to the normal or abnormal probability of each divided image grid.

The negative model inherits the identical backbone structure and is equipped with an independent detection head dedicated to defect classification. Its training dataset is composed of various defective samples, including broken yarns, dropped stitches and fly yarns, along with a large number of normal samples serving as interference data. Optimized by the combined BCE-Dice loss and Focal loss to address the challenges posed by tiny defects and imbalanced sample distribution, this model is designed to extract fine-grained features of similar subtle defects, categorize specific defect types, and suppress false positive detections triggered by complex meshes and three-dimensional structures of lace. It outputs the category, confidence score and position information of defects for each grid.

Regarding the training strategy, the two models are trained in a sequential manner: the positive model is trained first using positive samples, after which part of its backbone parameters are frozen to train the negative model based on the pre-trained positive model. All training experiments are conducted on a single NVIDIA RTX A4000 graphics card (NVIDIA Corporation, Santa Clara, CA, USA), with the total training duration ranging from 2 to 6 h.

For result fusion in inference, we define S_p as the abnormal probability output by the positive model and S_n as the defect confidence generated by the negative model. Two fixed thresholds $\tau_p = 0.5$ and $\tau_n = 0.6$ are adopted for industrial deployment. To be specific, the target area is judged as normal when $S_p < \tau_p$ and $S_n < \tau_n$; it is confirmed as a defective region with corresponding defect category when $S_p \geq \tau_p$ and $S_n \geq \tau_n$; and all other cases are marked for rechecking. The samples pending recheck will be highlighted on defect heatmaps for subsequent manual verification.

Dataset Partition and Splitting Principles

The dataset utilized in this research was gathered from real-world warp-knitted lace production lines located in Fujian Province, China. In total, 12,600 m of fabric were scanned with 28 RSJ and RSE series warp knitting machines, covering 112 kinds of lace patterns such as conventional mesh, 3D embossed and striped designs. The data collection process yielded 189,600 image frames, and each frame corresponds to a fabric area of 0.1 square meters. All raw images were divided into non-overlapping 128 × 128 pixel grids, generating 1,516,800 grid samples in total, among which the proportion of normal grids to defective grids is 94.7 to 5.3. This study defines five major defect categories for warp-knitted lace: broken yarns (BY), dropped stitches (DS), fly yarns (FY), misaligned yarns (MY) and pattern errors (PE). The overall number of defective grids reaches 79,990. Specifically, broken yarns account for 40.2% with 32,140 grids, dropped stitches take up 27.0% with 21,560 grids, fly yarns occupy 16.1% with 12,890 grids, misaligned yarns make up 12.2% with 9720 grids, and pattern errors correspond to 4.5% with 3690 grids. To comprehensively assess the generalization ability of the proposed model, we constructed an independent held-out test set consisting of 16 brand-new lace patterns that never appeared in the training and validation sets.

3.3.2. Model Architecture and Optimization

The detection backbone employs an improved YOLOv5s network integrated with the Convolutional Block Attention Module (CBAM) to enhance the attention to feature channels [24–26]. The multi-scale feature fusion layer adopts an FPN + PAN structure to strengthen the perception of defects with varying sizes [27]. A hybrid loss function combining BCE and Dice loss is adopted to effectively reduce missed detection of small-area defects [28]. The training environment is configured with an NVIDIA RTX A4000 GPU, with a single-card training duration of 2–6 h. The model size is approximately 85 MB, and the inference latency is less than 50 ms. To realize efficient deployment and on-site closed-loop application of artificial intelligence in weaving workshops, this system establishes a step-by-step model training and self-optimization system for lace fabric defect inspection. The model construction consists of two stages: the positive model resolves recognition bias caused by pattern differences between positive and negative samples and achieves accurate distinction between normal patterns and defective regions; the negative model focuses on addressing high similarity among abnormal samples and improves the ability to distinguish subtle defects from complex patterns. Through this dual-model collaboration mechanism, the system can complete model training and online deployment at the hour level, effectively adapting to the practical demands of numerous, frequently switched, and highly diverse patterns in lace production [29].

This system adopts a hybrid pipeline integrating grid-wise classification, anomaly detection and object detection as its overall task formulation. Grid-wise classification serves as the core task of the whole framework: the input raw images are partitioned into non-overlapping grids with a resolution of 128 × 128 pixels, and each grid is treated as a basic detection unit for independent classification. Specifically, every grid is first

identified as normal or defective, and defective grids are further categorized into specific defect types. The anomaly detection function is realized by the proposed positive-sample self-learning module, which distinguishes abnormal regions by calculating the feature deviation between real-time image grids and the pre-established normal feature space. This design enables the system to effectively identify subtle defects that share similar visual characteristics with normal lace patterns. Relying on the improved YOLOv5s network (Version v6.1, Ultralytics, Santa Clara, CA, USA, open-source software hosted on GitHub), the object detection module draws bounding boxes for confirmed defective regions and outputs the coordinate information and physical size of each defect. It is worth noting that this system does not conduct pixel-level semantic segmentation. Instead, defect heatmaps are generated according to the confidence scores of individual grids. To conclude, the entire detection framework is constructed as a hybrid solution combining grid-wise classification, anomaly detection and object detection.

In terms of core AI capability construction, the system forms six technical modules covering sampling, labeling, training, inference, localization, and self-learning. For efficient sampling, the system requires only 10–15 m of fabric samples to complete training data collection, with the full sampling process controlled within 15 min, enabling rapid data acquisition under small-sample conditions. For lightweight labeling, samples require no manual annotation, and the system supports a negative sample error tolerance rate of no more than 5%, significantly reducing manual participation and data preparation costs. For rapid training and deployment, the system supports model updating and online deployment at the hour level. Fast-learning models can be trained within 2–3 h and deep learning models optimized within 6 h on a single GPU, realizing instant adaptation to new patterns. For fine-grained inference, based on a single-grid inference mechanism, the system outputs multi-dimensional information, including defect coordinates, size, type, and confidence score, providing data support for fabric quality evaluation. For efficient localization, the system generates defect heatmaps to assist operators in locating defects accurately within 3 s, significantly improving manual review efficiency. For continuous self-learning, the system allows manual annotation and feedback on missed or falsely detected grids, and the model achieves self-optimization through incremental learning, continuously improving recognition accuracy and adaptability.

The defect heatmap adopted in this system is jointly generated by grid-level confidence scoring and GradCAM visual interpretation (open-source, GitHub, CAM-based interpretability tool developed by Facebook AI Research, Menlo Park, CA, USA), rather than relying on pure pixel-level segmentation outputs. The primary heatmap information is derived from the network's grid prediction results. Specifically, each 128×128 detection grid outputs a defect confidence score ranging from 0 to 1, and these continuous grid confidence values are mapped to a color gradient to construct an intuitive grid-level heatmap. To further enhance the visualization accuracy and highlight subtle defective regions, Grad-CAM visualization is incorporated as auxiliary supervision. The model's effective attention areas for potential defects are extracted and overlaid onto the original grid-based heatmap, which significantly strengthens the display capability of tiny and inconspicuous defects. Different from conventional segmentation-based heatmap generation, this fusion strategy combines prediction confidence distribution and model interpretability, achieving more reliable and targeted defect visualization for complex lace textures.

Figure 4 shows the algorithm detection flowchart. On the basis of algorithmic capabilities, the system further constructs three information-based and intelligent support functions for the weaving site. For intelligent hierarchical display, a three-level grid display strategy (red, yellow, green) is adopted to realize intuitive classification and visualization of defect severity, facilitating differentiated manual processing according to priority. For

automatic stop and defect control, through the Automatic Emergency Brake (AEB) and non-stop control mechanism of knitting machines, the "one operator with multiple machines" unattended production mode is realized, effectively reducing defect size and expansion. For automatic defect recording, the system supports automatic cutting, screening, and entry of defects, ensuring the integrity, accuracy, and real-time performance of defect information, and providing reliable data support for subsequent quality traceability and process optimization.

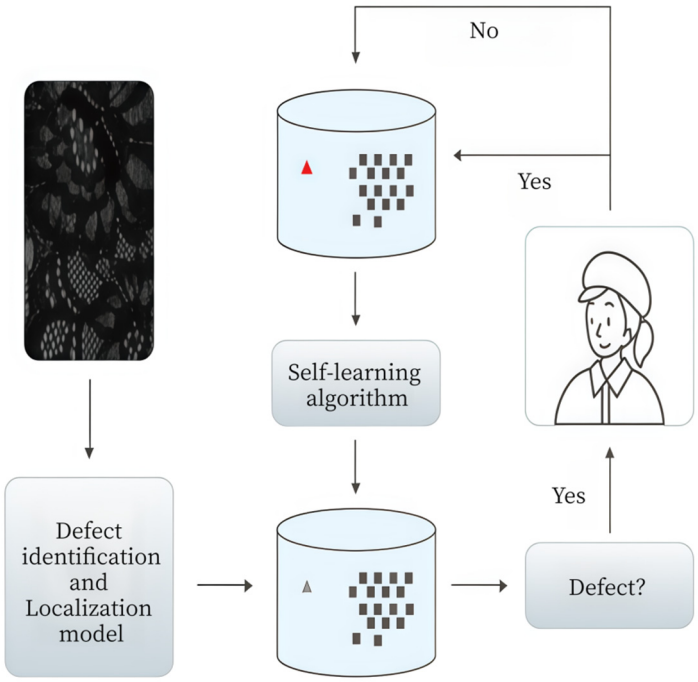


Figure 4. Flowchart of AI Algorithm Detection.

3.4. Client Function Design

To realize the digital and intelligent management of the production process in the weaving workshop, an intelligent client system integrating the weaving end, intermediate inspection end and finished product inspection end is constructed with AI defect detection as the core. By integrating AI and digital capabilities, the system achieves the fusion of detection results and logical rule judgment, releases early warning information accurately, and links alarm devices via PLC, thereby substantially improving the quality control and management efficiency of the weaving process [14,16].

Adopting a modular and hybrid architecture design, the client consists of the weaving terminal, intermediate inspection terminal and finished product inspection terminal, all uniformly connected to the server-side data center. The weaving terminal undertakes production monitoring and shutdown control; the intermediate inspection terminal is responsible for gray fabric detection and batch splitting management; and the finished product inspection terminal handles finished product inspection and report output. The system collects loom images and operational data through AI detection terminals, triggers shutdown or early warning via logical rule judgment, and synchronizes data to the server, realizing closed-loop management of production, inspection and statistics.

The weaving client is oriented to front-line production operators and integrates functions, including task configuration, defect identification, output statistics and AI learning. The system supports customized shutdown and defect entry rules by pattern, machine station or task, with built-in default rules to reduce repetitive configuration. The AI module conducts real-time analysis of fabric surface images and automatically shuts down the

machine and pushes alarm notifications when defects exceeding the threshold are detected. The interface displays machine status, output and qualification rate, and realizes visualized presentation of weaving status in combination with the workshop large-screen dashboard updated every 15 min. The one-click AI learning function enables operators to complete sample collection, model training and system deployment on the same platform, reducing operational complexity.

The intermediate inspection client is applied to the intermediate quality inspection of gray fabric, integrating human–machine collaboration and AI review functions. It supports automatic merging of long defects and damage loss calculation to simplify entry operations, and enables piece splicing and barcode printing by roll number, replacing manual handwriting with machine printing. The re-inspection process is initiated by the server and executed by the client to ensure task closed-loop management. Output statistics are calculated by team average allocation to enhance the fairness of performance assessment. Users can also register structural parameters such as pattern height and stripe width to meet the requirements of traceability and data analysis.

The finished product inspection client serves the finished product inspection process, supporting code scanning task execution, AI-assisted detection and label printing. During inspection, the AI automatically identifies latent defects such as inconsistent pattern height and prompts operators for correction. The system generates task records and output data based on inspection results, and the printed labels contain parameters, including pattern model, customer information, length, width and weight. After data upload, the server automatically generates inspection details and quality reports, which support the export of customer-version reports and the summary of personnel performance.

System anomalies are classified into event type and early warning type. Event-type anomalies cover operational nodes such as process acceptance, equipment maintenance, beam replacement and fabric doffing, with automatic recording of initiation, response and completion time. For early warning-type anomalies, threshold values are set for indicators, including qualification rate, shutdown frequency, operation rate and maintenance duration; once thresholds are exceeded, alerts are pushed to corresponding management levels such as team leaders, workshop directors and factory directors. WeChat Official Accounts and DingTalk groups are adopted as push channels to achieve real-time response and multi-level linkage for abnormal events [14,16].

Each client automatically generates data reports upon task completion, including team output statements, individual output statements, fabric inspection detailed sheets and quality reports, realizing data visualization and full traceability. Since the system operation, the collaboration of AI detection and logical judgment has significantly improved the accuracy of weaving defect identification and production response speed; data reporting and the early warning mechanism have strengthened quality management and performance appraisal. The system was exhibited and applied at the 7th Digital China Summit and received media coverage, verifying its popularization value in the digital transformation of textile manufacturing.

4. System Implementation Effect

Centering on the key technical objectives of the AI visual quality inspection system for warp-knitted lace, systematic technical research is conducted from the aspects of computing power platform construction, algorithm model optimization, and detection accuracy improvement. Through collaborative design of software and hardware, algorithm innovation, and data-driven self-learning mechanism, the system has achieved comprehensive improvement in overall performance, detection accuracy and application stability.

4.1. Construction of AI Computing Power Platform and Improvement of System Performance

4.1.1. Configuration of Ascend Computing Cluster

To satisfy the high computational demands of AI algorithms, this system deploys the Huawei Atlas intelligent computing solution based on Ascend series AI chips (Huawei Technologies Co., Ltd., Shenzhen, Guangdong, China). This integrated architecture covers terminal, edge and cloud layers, and supports the full workflow of deep learning models from offline training to online inference. Powered by the self-developed Da Vinci architecture, Ascend processors exhibit excellent computing performance and energy efficiency. The main processor provides FP16 computing power of 256–320 TFLOPS and INT8 computing power of 512–640 TOPS with a power consumption of merely 310 W. As a lightweight inference chip, Ascend 310 delivers 16 TOPS of INT8 performance while consuming only 8 W (Huawei Technologies Co., Ltd., Shenzhen, Guangdong, China), which significantly boosts the operational efficiency of field deployment.

The Atlas training server is configured with eight Ascend AI processors, providing a total FP16 computing power of 2.24 PFLOPS. Each NPU is connected via three HCCS high-speed interconnection channels with a maximum bandwidth of 90 GB/s, enabling low-latency data interaction and resource sharing across multiple chips. Equipped with PCIe 4.0 and RoCE v2 interfaces, the platform supports both horizontal expansion and vertical performance upgrading, which greatly enhances system scalability and concurrent processing capacity [30–32]. Advanced heat dissipation design and dynamic power management are also applied to guarantee stable operation under high-load computing conditions. After overall optimization, the system's comprehensive computing capability is raised by three times. Correspondingly, the fabric inspection speed is increased from 10 m/min in traditional manual inspection to 30 m/min, which achieves a substantial improvement in detection efficiency and provides reliable hardware support for large-scale industrial application [33].

4.1.2. Dual Hardware Platform Division for Training and Deployment

Two sets of hardware platforms are configured for model training and industrial deployment respectively, and all tests are conducted on corresponding devices to guarantee fair and standardized evaluation. The offline training platform adopts an NVIDIA RTX A4000 GPU with 8 GB video memory, which undertakes model training, data augmentation and automatic label generation. The complete training cycle of the proposed model ranges from 2 to 6 h, and its single-frame inference latency is 42 ms. All algorithm evaluation indicators, including accuracy, F1-score, mAP and results of ablation experiments, are tested uniformly on this platform. For on-site industrial scenarios, the system adopts Huawei Ascend 310 chip matched with the Atlas AI server to realize edge inference, 5G data transmission and production equipment linkage. The single-frame inference latency on this deployment platform is 38 ms, and its computing performance is three times higher than that of conventional industrial control hosts. All field indicators such as real-time detection speed, equipment linkage response and long-term operational stability are collected based on this industrial deployment platform.

4.2. AI Algorithm Optimization and Model Robustness Enhancement

Aiming at the complex textures, diverse patterns and subtle chromatic aberration of warp-knitted lace, this study develops a dual-path AI detection algorithm combining classification and semantic segmentation. The proposed algorithm realizes defect location, category identification and severity assessment, and provides reliable decision support for automatic loom control in actual production [34]. The grid-level comprehensive classification accuracy of the system reaches 95%.

In the training phase, Label Smoothing and a multi-loss-based cross-validation mechanism are introduced to address noisy and incorrect labels in traditional annotation workflows. Conventional hard labels tend to cause model overfitting and loss of inter-class semantic information in multi-classification tasks. By converting deterministic labels into probabilistic distribution via the smoothing parameter ε, soft labels preserve inter-class similarity and data semantic structures, lower the model's sensitivity to annotation errors, and greatly improve generalization performance. To further strengthen model robustness, we adopt joint training with multiple loss functions, including Cross Entropy Loss, Focal Loss, SoftTriple Loss and Asymmetric Loss [35]. This design enables the model to handle hard samples and imbalanced datasets stably. After iterative optimization and validation, the algorithm is deployed for on-site lace defect recognition and linked with looms. The system can automatically trigger alarm or shutdown operations according to defect types and severity, forming a closed-loop workflow of intelligent detection, automatic decision and equipment response, which effectively improves production automation and quality management level.

4.2.1. Quantitative Evaluation Metrics

To quantitatively evaluate the overall detection and classification performance, we adopt a set of mainstream computer vision evaluation metrics. Before presenting specific test results, we clarify the definitions of all indicators uniformly to avoid ambiguity in subsequent analysis. Evaluated on the standard test set, the proposed method achieves an mAP@0.5 of 94.2%, overall precision of 95.1%, overall recall of 96.3% and overall F1-score of 95.7%. The corresponding false positive rate (FPR) and false negative rate (FNR) are 1.8% and 1.2%, respectively.

We further conduct category-wise evaluation on the five predefined defect types. Broken yarns achieve the best performance, with precision, recall and F1-score of 96.4%, 97.1% and 96.7%. The three metrics for dropped stitches are 95.2%, 96.5% and 95.8%; those for fly yarns are 94.7%, 95.8% and 95.2%; and those for misaligned yarns are 93.9%, 94.6% and 94.2%. Pattern errors obtain the lowest performance across all categories, with precision of 92.1%, recall of 93.2% and F1-score of 92.6%. An additional test is carried out on an independent held-out set composed of unseen lace patterns to verify model generalization. On this dataset, the model yields a precision of 92.7%, recall of 93.5%, F1-score of 93.1%, FPR of 2.9% and FNR of 2.1%. We also analyze the confusion matrix covering six classes (one normal class and five defect classes). Results show that misclassification mainly occurs between fly yarns and misaligned yarns due to their similar visual features. A small number of normal texture grids are mistakenly identified as pattern errors, while the overall classification error remains at a low level.

All core indicators are defined clearly as follows. The 95% classification accuracy mentioned throughout the paper refers to grid-level comprehensive classification accuracy. Calculated on all 128×128 pixel grid samples, this metric represents the ratio of correctly classified grids (normal or defective) to the total number of grids, with the formula. The frame-level automatic detection rate, which exceeds 98%, is defined as the defect hit rate for fabric image frames. For frames containing real defects, the system can successfully locate abnormal regions in over 98% of cases, reflecting the practical defect capture capability in production. For other standard metrics, FPR refers to the proportion of normal grids misjudged as defective among all normal samples; FNR represents the proportion of real defective samples that are not detected among all defective samples. mAP@0.5 denotes the mean average precision calculated with an IoU threshold of 0.5 for object detection tasks.

4.2.2. Ablation and Baseline Comparison Analysis

We compare the proposed method with multiple mainstream algorithms to validate its technical advantages. The comparison group covers traditional machine vision methods, dedicated textile defect detectors, classic object detection models and modern anomaly detection algorithms. We select GLCM + Gabor-based traditional machine vision methods, professional textile detectors, vanilla YOLOv5s, YOLOv8n and autoencoder (AE)-based anomaly detection models as baselines. Table 2 summarizes the quantitative results of all methods in terms of precision, recall, F1-score, FPR and FNR.

Table 2. The quantitative results of all methods.

Method	Precision	Recall	F1-Score	FPR	FNR
Classical MV (GLCM + Gabor)	72.3%	70.1%	71.2%	8.7%	9.5%
Textile traditional detector	78.6%	77.2%	77.9%	6.2%	7.1%
Vanilla YOLOv5s	86.5%	87.2%	86.8%	4.3%	3.8%
YOLOv8n	89.1%	89.7%	89.4%	3.1%	2.6%
Modern Anomaly Detection (AE)	90.2%	91.0%	90.6%	2.5%	2.3%
Proposed Method	95.1%	96.3%	95.7%	1.8%	1.2%

Traditional machine vision methods rely on handcrafted features and fail to adapt to complex textures and diverse patterns of warp-knitted lace, thus presenting the poorest performance and high error rates. Traditional textile detectors achieve moderate results but cannot effectively distinguish subtle defects with similar appearances. Although vanilla YOLOv5s and YOLOv8n outperform conventional methods, they lack targeted optimization for tiny low-contrast defects and are vulnerable to complex background interference. The AE-based anomaly detection method has good robustness but cannot realize fine-grained defect classification and suffers from insufficient feature discrimination for similar texture defects.

Different from the above methods, our algorithm integrates positive-sample self-learning (PSSL), dual-model collaboration and multi-loss joint optimization. It effectively suppresses false alarms caused by intricate lace textures and strengthens feature extraction for subtle defects. The proposed method outperforms all baselines on all metrics, which fully verifies the superiority and effectiveness of the designed framework.

4.2.3. Ablation Study of Core Modules and Mechanisms

To quantify the contribution of each functional module, we conduct step-by-step ablation experiments. All tests adopt F1-score as the unified evaluation metric and are implemented under identical experimental conditions, including dataset partition, training hyperparameters and hardware environment. The tested modules include the CBAM attention mechanism, FPN + PAN multi-scale fusion structure, BCE-Dice hybrid loss function, self-supervised learning strategy, positive-sample self-learning (PSSL), single negative model and dual-model collaboration mechanism. The ablation results are listed in Table 3.

As shown in the table, each module brings steady performance improvement on the original baseline. The CBAM attention module enhances feature perception for subtle defects, and the FPN + PAN structure optimizes multi-scale feature fusion for complex lace textures. The BCE-Dice hybrid loss alleviates training bias brought by imbalanced

samples and tiny target sizes. Combined with self-supervised learning and PSSL, the model reduces dependence on labeled defective samples and better learns the feature distribution of normal lace textures. The introduced negative model further suppresses false positives induced by interference textures. The complete framework integrating all modules and dual-model collaboration achieves the highest F1-score. It can be concluded that positive-sample self-learning and dual-model collaboration are the two most critical components, which contribute the most to performance improvement and guarantee high-precision and robust detection of subtle defects in warp-knitted lace.

Table 3. The ablation results.

Model Configuration	F1-Score
Vanilla YOLOv5s (Baseline)	86.8%
+ CBAM	89.2%
+ CBAM + FPN + PAN	91.5%
+ CBAM + FPN + PAN + BCE-Dice loss	92.8%
Above + Self-supervised learning	94.1%
Above + Positive-sample self-learning (PSSL)	94.9%
Above + Single Negative Model	95.2%
Full model (All modules + Dual-model collaboration)	95.7%

4.3. Self-Supervised Learning and Detection Accuracy Improvement

To break the reliance of traditional supervised learning on annotated data, this study constructs a self-supervised training system based on massive fabric samples. It enhances the general representation capability of the model through feature learning from unlabeled data. With the encoder as the core, the system realizes multi-scale feature extraction via feature-level and image-level reconstruction, thus exhibiting higher sensitivity and accuracy in subtle defect detection.

Three key technologies are introduced into the deep neural network structure, namely Dropout Layer, Noisy Bottleneck and Layer Group shared computing structure [36,37]. Dropout Layer breaks the co-adaptation between network layers by randomly discarding partial neuron outputs, which effectively alleviates overfitting and improves model generalization [38].

Noisy Bottleneck destroys excessive feature dependence through noise injection, enabling the model to learn more stable and robust feature expressions and enhancing the detection capability for local anomalies.

The Layer Group sharing structure optimizes training efficiency through layered sharing of computing resources and reduces computational redundancy, ensuring stable model performance in scenarios with multiple products and complex textures.

Furthermore, a Loose Loss Reconstruct mechanism is proposed in this study. By controlling the learning dependency between network layers, it avoids overfitting in single-category feature learning and improves the model's generalization across different textures and patterns.

In terms of model interpretability and visual analysis, GradCAM heatmap and t-SNE semantic feature visualization methods are adopted (Inria, Saclay, France). Figure 5 presents the lace defect images, and Figure 6 shows the corresponding heatmaps. Grad-CAM visualizes the model's attention regions and reveals the distribution of key features for classification decision-making; t-SNE dimensionality reduction displays the semantic clustering relationship of different categories, facilitating the identification of potential

misannotated samples and hard-to-classify samples. Meanwhile, white-box analysis tools assist researchers in understanding the model's decision logic and provide a reference for subsequent algorithm optimization.

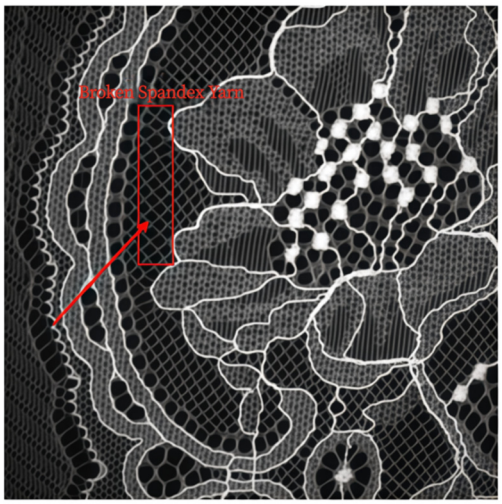


Figure 5. Lace defect images.

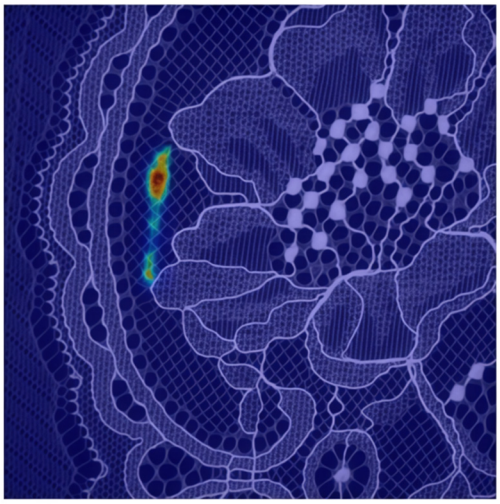


Figure 6. Corresponding heatmaps.

Through systematic technical research, the detection accuracy of the AI quality inspection system has been remarkably improved. The recognition precision of lace defects reaches 1 mm, which can accurately capture single-yarn-level anomalies of 90D yarn. The accuracy of AI defect classification is increased from the initial 40% to 95%, and the overall automatic detection rate exceeds 98% [17,39,40].

This section analyzes typical detection results and challenging failure cases of the proposed model on warp-knitted lace defect detection In terms of true positive detection, the model achieves reliable performance on common and obvious defects such as broken yarns and dropped stitches, which can be accurately localized and correctly categorized by the system. The primary source of false positive errors originates from the complex structural characteristics of lace: individual intricate mesh textures and three-dimensional embossed surfaces of normal lace are highly visually similar to fly yarn defects, resulting in occasional misclassification of normal texture regions as defective areas. For false negative cases, detection failures mainly occur on ultra-fine single-yarn deviation defects. Such tiny defects are close to the 90D yarn fineness limit and present extremely subtle feature differences, especially under dark lighting conditions, making them difficult to be identified

by the model. In addition, we summarize three typical difficult detection scenarios that restrict further performance improvement, including subtle defects with visual features almost indistinguishable from normal lace patterns, pattern errors located at fabric edges with incomplete texture information, and overlapping tiny defects of multiple categories with confused feature representations. These challenging cases reflect the core difficulties of fine-grained lace defect detection and also provide effective guidance for subsequent model optimization and algorithm improvement.

4.4. Industrial On-Site Validation Details

We carried out a 180-day continuous field deployment to fully verify the practical performance, economic benefits and operational reliability of the proposed intelligent detection system. The system was applied to 28 warp-knitted lace looms for full-process online inspection, with a total inspected fabric length of 89,200 m across various production batches. The large-scale field data effectively prove the reliability and generalization of the system. In terms of manpower arrangement, the original all-manual inspection required 56 on-site operators, while the AI system cuts the number down to 22. Each operator can manage multiple looms simultaneously to improve working efficiency. During the whole validation period, the system identified 12,478 real fabric defects, and the false-stop rate remained as low as 1.1%. This metric refers to unintended loom shutdowns triggered by false detection alarms, reflecting the system's high detection precision and little disruption to normal production.

The long-term field data also reveal prominent improvements in product quality and labor cost control. The average fabric reject rate dropped from 7.2% under manual inspection to 5.04% after system deployment, a relative reduction of 30% calculated by $\frac{7.2\%-5.04\%}{7.2\%} \times 100\%$. The headcount of frontline inspectors fell from 56 to 22, equivalent to a direct staff reduction rate of about 60.7% based on $\frac{56-22}{56} \times 100$. Taking auxiliary management roles into account, the overall labor cost for fabric inspection was reduced by 40%, which is consistent with the economic benefit described in the abstract.

We further evaluated system reliability using two key indicators: Mean Time Between Failures (MTBF) and software stability rate. System failures are defined as image acquisition errors, model inference crashes, loom linkage malfunctions and persistent abnormal alarms. According to field statistics, the system achieved an MTBF of more than 900 h. Its software stability rate, calculated as the ratio of normal running time to total running time, reached 99.8%. All results demonstrate that the system is capable of stable, long-term intelligent detection in actual warp-knitted lace production lines.

4.5. Application Boundary and Limitations

To clarify the applicable scope and practical limitations of the proposed detection system, the implementation boundary conditions under actual industrial scenarios are summarized in this section. In terms of material adaptability, the model is primarily designed and optimized for warp-knitted lace with yarn fineness ranging from 90D to 150D, which constitutes the main production object of this study. For yarn fineness below 90D or above 150D, the detection accuracy will decrease by 3–6% due to mismatched texture feature scales. Regarding pattern adaptability, the system achieves stable detection performance for conventional mesh, striped, and three-dimensional embossed lace types, while the false positive rate will rise to 4–5% when facing ultra-complex artistic jacquard lace with highly irregular textures. For environmental lighting constraints, the system maintains reliable detection under standard factory indoor lighting with an illuminance of 300–800 lux; however, extremely low-light environments below 150 lux or strong reflected light exceeding 1000 lux will cause an accuracy drop of 4–7%. In terms of production line

adaptation, the developed system is well compatible with RSJ and RSE series warp-knitted looms deployed in Fujian industrial sites. For ultra-high-speed looms with a running speed exceeding 40 m/min, further parameter tuning and adaptive optimization are required to guarantee qualified real-time detection performance.

5. Conclusions

This study targets the intelligent upgrading requirements of the warp-knitted lace industry and develops a 5G-based AI visual defect inspection system. Combining machinery, electronics, optics, computing and software technologies, the platform builds a full-process digital closed-loop system that integrates image acquisition, intelligent identification, logic judgment, alarm notification and data management. Different from traditional defect detection methods relying on massive defective samples and handcrafted features, this work proposes a positive-sample self-learning paradigm and a dual-model collaboration framework. Equipped with optimized network modules and multi-loss joint training strategies, the solution effectively addresses the core challenges of complex texture interference and subtle defect recognition in lace production.

Field deployment proves that the developed system delivers outstanding comprehensive performance in detection accuracy, operational efficiency and long-term stability. It greatly cuts manual inspection workload and product defective rate, and maintains a reliable running status in actual production environments. This research realizes the engineering transformation of AI vision inspection technology from laboratory research to large-scale industrial application. It not only breaks through the technical bottlenecks of traditional manual inspection for warp-knitted lace but also provides a universal, replicable intelligent transformation scheme for the whole textile industry, which has prominent practical value and promotional significance. According to the application boundary and limitations summarized in the industrial test, follow-up research will carry out targeted optimization and expansion.

Author Contributions: T.Y. conceived the overall research idea and designed the systematic framework of this study. Y.W. and Y.S. conducted the comprehensive experiments, data collation and quantitative analysis. C.L. and L.J. participated in system deployment, engineering implementation and algorithm optimization. Y.S. supervised the entire research work, managed the project progression and revised the manuscript. All authors have read and agreed to the published version of the manuscript.

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References

1. Orisadare, E.A.; Achukwu, O.E.; Ogunyemi, A.A.; Adedeji, D.O.; Diyaolu, I.J.; Ugwu, E.I.; Oluwatope, A.O.; Bakare, K.O. Digitalisation and Green Strategies: A Systematic Review of the Textile, Apparel and Fashion Industries. *Circ. Econ. Sustain.* **2025**, *5*, 2775–2827. [CrossRef]
2. Rasheed, A.; Zafar, B.; Rasheed, A.; Ali, N.; Sajid, M.; Dar, S.H.; Habib, U.; Shehryar, T.; Mahmood, M.T. Fabric Defect Detection Using Computer Vision Techniques: A Comprehensive Review. *Math. Probl. Eng.* **2020**, *2020*, 8189403. [CrossRef]
3. Hanbay, K.; Talu, M.F.; Özgüven, Ö.F. Fabric defect detection systems and methods—A systematic literature review. *Optik* **2016**, *127*, 11960–11973. [CrossRef]
4. Kahraman, Y.; Durmuşoğlu, A. Deep learning-based fabric defect detection: A review. *Text. Res. J.* **2022**, *93*, 004051752211307. [CrossRef]
5. Machado, R.; Barros, L.A.; Vieira, V.; Silva, H.C.; Carvalho, V. Textile Defect Detection Using Artificial Intelligence and Computer Vision—A Preliminary Deep Learning Approach. *Electronics* **2025**, *14*, 3692. [CrossRef]
6. Kumar, A. Computer-Vision-Based Fabric Defect Detection: A Survey. *IEEE Trans. Ind. Electron.* **2008**, *55*, 348–363. [CrossRef]
7. He, F.; Xin, B.; Zhan, Z.; Yang, W.; Liu, J. A review of fabric defect detection in textile manufacturing. *J. Text. Inst.* **2025**, *117*, 327–345. [CrossRef]
8. Xie, G.; Xu, Y.; Yu, Z.; Sun, Y. An intelligent defect detection system for warp-knitted fabric. *Text. Res. J.* **2021**, *92*, 1394–1404. [CrossRef]
9. Mei, S.; Wang, Y.; Wen, G. Automatic Fabric Defect Detection with a Multi-Scale Convolutional Denoising Autoencoder Network Model. *Sensors* **2018**, *18*, 1064. [CrossRef] [PubMed]
10. Zhang, Y.; Shi, W.; Zhang, J. Detection of Defects in Warp Knitted Fabrics Based on Local Feature Scale Adaptive Comparison. *Appl. Sci.* **2024**, *14*, 10754. [CrossRef]
11. Ma, S.; Liu, Y.; Zhang, Y. Fabric Defect Detection Based on Improved Lightweight YOLOv8n. *Appl. Sci.* **2024**, *14*, 8000. [CrossRef]
12. Chu, Y.H.; Feng, D.Q.; Liu, Z.Z.; Zhao, Z.Z.; Wang, Z.Z.; Xia, X.G.; Quek, T.Q.S. Hybrid-learning-based operational visual quality inspection for edge-computing-enabled IoT system. *IEEE Internet Things J.* **2022**, *9*, 4958–4972. [CrossRef]
13. Coll-Perales, B.; Gozalvez, J.; Maestre, J.L. 5G and beyond: Smart devices as part of the network fabric. *IEEE Netw.* **2019**, *33*, 170–177. [CrossRef]
14. Xu, Y.; Sun, R.; Zhi, C.; Liu, Z.; Chen, J.; Ke, Z.; Yu, L. Zero-defect manufacturing in the textile industry: A review of current advances and challenges. *Text. Res. J.* **2024**, *95*, 194–219. [CrossRef]
15. Xu, C.Q.; Xu, L.C.; Luo, K.; Zhang, J.T.; Sitahong, A.; Yang, M.; Zhang, C.J. The essence and applications of machine vision inspection for textile industry: A review. *J. Text. Inst.* **2025**, *116*, 2286–2310. [CrossRef]
16. Varga, P.; Peto, J.; Franko, A.; Balla, D.; Haja, D.; Janky, F.; Soos, G.; Ficzere, D.; Maliosz, M.; Toka, L. 5G support for Industrial IoT Applications—Challenges, Solutions, and Research gaps. *Sensors* **2020**, *20*, 828. [CrossRef] [PubMed]
17. Kreuzberger, D.; Kühl, N.; Hirschl, S. Machine Learning Operations (MLOps): Overview, Definition, and Architecture. *IEEE Access* **2023**, *11*, 31866–31879. [CrossRef]
18. Nahar, L.; Awrangjeb, M.; Islam, M.S. AI-enabled defect detection in industrial products: A comprehensive survey, key insights and future research challenges. *Adv. Eng. Inform.* **2025**, *69*, 104067. [CrossRef]
19. Yu, H.; Li, D.; Chen, Y. A state-of-the-art review of image motion deblurring techniques in precision agriculture. *Heliyon* **2023**, *9*, e17332. [CrossRef] [PubMed]
20. Ren, Z.; Fang, F.; Yan, N.; Wu, Y. State of the Art in Defect Detection Based on Machine Vision. *Int. J. Precis. Eng. Manuf.-Green Technol.* **2021**, *9*, 661–691. [CrossRef]
21. Di, L.; Long, H.; Liang, J. Fabric Defect Detection Based on Illumination Correction and Visual Salient Features. *Sensors* **2020**, *20*, 5147. [CrossRef] [PubMed]
22. Oh, W.; In, C.; Oh, Y.; Cho, K. Effects of PWM Dimming LED Illumination on Camera Images and Countermeasures. In Proceedings of the 2019 IEEE 60th International Scientific Conference on Power and Electrical Engineering of Riga Technical University (RTUCON), Riga, Latvia, 7–9 October 2019; pp. 1–7. [CrossRef]
23. Mumuni, A.; Mumuni, F. Data augmentation: A comprehensive survey of modern approaches. *Array* **2022**, *16*, 100258. [CrossRef]
24. Lin, G.; Liu, K.; Xia, X.; Yan, R. An Efficient and Intelligent Detection Method for Fabric Defects based on Improved YOLOv5. *Sensors* **2022**, *23*, 97. [CrossRef] [PubMed]
25. Woo, S.; Park, J.; Lee, J.-Y.; Kweon, I.S. CBAM: Convolutional Block Attention Module. In *Computer Vision—ECCV 2018*; Springer: Berlin/Heidelberg, Germany, 2018; pp. 3–19. [CrossRef]
26. Wang, Y.; Xu, Y.; Yu, Z.; Xie, G. Color-patterned fabric defect detection based on the improved YOLOv5s model. *Text. Res. J.* **2023**, *93*, 4792–4803. [CrossRef]
27. Lin, T.-Y.; Dollar, P.; Girshick, R.; He, K.; Hariharan, B.; Belongie, S. Feature Pyramid Networks for Object Detection. In 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). In Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, 21–26 July 2017. [CrossRef]

28. Bai, D.; Li, G.; Jiang, D.; Yun, J.; Tao, B.; Jiang, G.; Sun, Y.; Ju, Z. Surface defect detection methods for industrial products with imbalanced samples: A review of progress in the 2020s. *Eng. Appl. Artif. Intell.* **2024**, *130*, 107697. [CrossRef]
29. Yang, C.; Wang, Y.; Lan, S.; Wang, L.; Shen, W.; Huang, G.Q. Cloud-edge-device collaboration mechanisms of deep learning models for smart robots in mass personalization. *Robot. Comput.-Integr. Manuf.* **2022**, *77*, 102351. [CrossRef]
30. Gangidi, A.; Miao, R.; Zheng, S.; Bondu, S.J.; Goes, G.; Morsy, H.; Puri, R.; Riftadi, M.; Shetty, A.J.; Yang, J.; et al. RDMA over Ethernet for Distributed Training at Meta Scale. In Proceedings of the ACM SIGCOMM 2024 Conference, Sydney, NSW, Australia, 4–8 August 2024; pp. 57–70. [CrossRef]
31. Li, A.; Song, S.L.; Chen, J.; Li, J.; Liu, X.; Tallent, N.R.; Barker, K.J. Evaluating Modern GPU Interconnect: PCIe, NVLink, NV-SLI, NVSwitch and GPUDirect. *IEEE Trans. Parallel Distrib. Syst.* **2019**, *31*, 94–110. [CrossRef]
32. Sapio, A.; Canini, M.; Ho, C.-Y.; Nelson, J.; Kalnis, P.; Kim, C.; Krishnamurthy, A.; Moshref, M.; Ports, D.R.K.; Richtarik, P.; et al. Scaling Distributed Machine Learning with In-Network Aggregation. *arXiv* **2019**, arXiv:1903.06701. [CrossRef]
33. Barman, J.; Wu, H.-C.; Kuo, C.-F.J. Development of a real-time home textile fabric defect inspection machine system for the textile industry. *Text. Res. J.* **2022**, *92*, 4778–4788. [CrossRef]
34. Czimmermann, T.; Ciuti, G.; Milazzo, M.; Chiurazzi, M.; Roccella, S.; Oddo, C.M.; Dario, P. Visual-Based Defect Detection and Classification Approaches for Industrial Applications—A SURVEY. *Sensors* **2020**, *20*, 1459. [CrossRef] [PubMed]
35. Ridnik, T.; Ben-Baruch, E.; Zamir, N.; Noy, A.; Friedman, I.; Protter, M.; Zelnik-Manor, L. Asymmetric Loss For Multi-Label Classification. In Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV), Montreal, QC, Canada, 10–17 October 2021. [CrossRef]
36. Alzubaidi, L.; Zhang, J.; Humaidi, A.J.; Al-Dujaili, A.; Duan, Y.; Al-Shamma, O.; Santamaría, J.; Fadhel, M.A.; Al-Amidie, M.; Farhan, L. Review of deep learning: Concepts, CNN architectures, challenges, applications, future directions. *J. Big Data* **2021**, *8*, 53. [CrossRef] [PubMed]
37. Khan, A.; Sohail, A.; Zahoora, U.; Qureshi, A.S. A survey of the recent architectures of deep convolutional neural networks. *Artif. Intell. Rev.* **2020**, *53*. [CrossRef]
38. Ait Skourt, B.; El Hassani, A.; Majda, A. Mixed-pooling-dropout for convolutional neural network regularization. *J. King Saud. Univ.-Comput. Inf. Sci.* **2021**. [CrossRef]
39. Carrilho, R.; Yaghoubi, E.; Lindo, J.; Hambarde, K.; Proença, H. Toward Automated Fabric Defect Detection: A Survey of Recent Computer Vision Approaches. *Electronics* **2024**, *13*, 3728. [CrossRef]
40. Beljadid, A.; Tannouche, A.; Balouki, A. Automatic fabric defect detection employing deep learning. *Int. J. Electr. Comput. Eng.* **2022**, *12*, 4129–4136. [CrossRef]

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